

Subham Karmakar

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EDUCATION

Adamas University

B.Tech in Computer Science and Engineering | Kolkata, West Bengal, India

Aug 2024 – May 2028 (Expected)

CGPA: 9.17/10

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, C, C++, SQL

Frontend: React.js, Next.js, Tailwind CSS, Redux Toolkit, GSAP, Framer Motion, HTML5/CSS3

Backend: Node.js, Express.js, FastAPI, REST API, WebSockets, Socket.io, Stripe

Auth & Security: JWT, OAuth, RBAC, Session Management

Databases: PostgreSQL, pgvector, MongoDB, Redis, SQLite, Supabase, Neon

Tools & DevOps: Git, Docker, Docker Compose, GitHub Actions, Nginx, PM2, Vercel, Render, Sentry, Linux, CI/CD

Testing: Jest, Vitest, pytest

WORK EXPERIENCE

Adamas University

Nov 2025 – Present

Moodle LMS Cloud Maintainer (Part-time)

Kolkata, India

- Maintained 99%+ uptime for 1,000+ students and faculty by administering cloud-hosted Moodle LMS with proactive health monitoring and incident response workflows.
- Reduced faculty support tickets by ~40% and per-course setup time by ~half by redesigning role-permission flows and standardizing plugin configuration for 100+ faculty members.

Ryze AI

Jan 2026 – Apr 2026

Frontend Developer Intern

Remote

- Accelerated feature releases by shipping production UI components (React.js, TypeScript, Tailwind CSS) and participating in code reviews as author and reviewer — zero missed sprint commitments across the engagement.
- Reduced average data-fetch latency by ~30% by integrating REST API endpoints with optimized request patterns and component-level caching strategies.

PROJECTS

CyCoder | *Next.js, TypeScript, PostgreSQL, Docker, Nginx, PM2* | cycoders.in

2025

- Engineered JWT-based multi-role auth (Admin/Member) with RBAC access controls and QR-code check-in, scaling Adamas developer community to 100+ active users via gamified XP-progression and leaderboards on a Next.js + PostgreSQL stack.
- Achieved zero-downtime deployments by containerizing the full TypeScript stack with Docker Compose, VPS-hosted behind Nginx with PM2 and GitHub Actions CI/CD pipeline.

Noted | *Next.js, TypeScript, Node.js, WebSockets, PostgreSQL, Redis, Stripe, LLM APIs* | notezero.vercel.app

2025

- Built a real-time collaboration engine with zero data loss under concurrent multi-user edits via WebSocket pub/sub over Redis with operational-transform-inspired conflict resolution — client state managed with Redux Toolkit.
- Reduced note organization effort by ~95% with an agentic LLM pipeline (intent classification, tag generation, context-aware linking); integrated Stripe for subscription billing.

Web-Server | *TypeScript, Node.js* | github.com/subham12r/web-server

2026

- Built a production-grade HTTP/1.1 server from raw TCP sockets in TypeScript — full request/response parsing, keep-alive connection reuse, streaming BodyReader, and explicit socket back-pressure; demonstrates deep understanding of the networking layer underlying all RESTful API design.

Panda | *Electron 35, React 19, TypeScript, Python FastAPI, SQLite, fastembed*

2026

- Built a local-first AI desktop assistant with multi-model support (Anthropic/Ollama/Groq), session-scoped RAG data ingestion pipeline (sentence-transformer embeddings, top-5 chunk injection), agentic tool-orchestration research pipeline, and WebSocket streaming — 100% local.

CrackCoded | *Next.js, TypeScript, FastAPI, Python, PostgreSQL, pgvector* | crackcoded.vercel.app

2026

- Delivered 340% faster learning and 67% lower time-per-problem via pgvector SM-2 spaced-repetition queues and an AI hint engine; 72% average topic completion rate — Python/FastAPI backend deployed on Render.

ACHIEVEMENTS & CERTIFICATIONS

Claude Code 101 & MCP Introduction — Anthropic Education, Jun 2026 | skilljar/y7qppxfzwffs | skilljar/s4t3vecwu84e

Smart India Hackathon 2025 — Internal Finalist, Top 10 in Software Category out of 150 teams

Certified Full Stack Web Developer — Udemy, 97 hours (ID: UC-f427f3a7-351a-41fc-91e4-510256962222)